BTS–VI(SS)–01.19–0033 Reg. No. [ ]

A

B.Tech. Degree VI Semester Special Supplementary Examination January 2019

ME 15-1605 CAD/CAM
(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) Explain briefly the concept of constructive solid geometry (CSG) in geometric modelling.
(b) What is Detroit type of automation?
(c) List the different types of NC words with examples.
(d) Differentiate between generative verses variant types of Computer aided Process Planning.
(e) Distinguish between (i) fixed zero & floating zero (ii) absolute positioning and incremental positioning.
(f) List four essential requirements of an actuating mechanism in CNC.
(g) What is machining center?
(h) List the three major classification of applications of Robots.
(i) What is lead through teaching of Robot?
(j) What is Computer Integrated Manufacturing System (CIMS).

PART B

(4 × 10 = 40)

II. (a) Discuss briefly the four basic elements in CAD: Geometric modelling/Database – Engineering Analysis – Design review evaluation – Automatic Drafting. (6)
(b) Explain following elements in line balancing problem: (4)
   (i) Minimum Rational work element
   (ii) Total work content
   (iii) Workstation process time
   (iv) Cycle time

OR

III. (a) Explain steps in data exchange between CAD and CAM. (4)
(b) A 30 station transfer line is being proposed to machine a certain component currently produced by conventional methods. The proposal received from the machine tool builder states that the line will operate at 100% efficiency at a production rate of 100 pc/hr. From a similar transfer line it is estimated that breakdowns of all types will occur at a frequency of F=0.20 breakdowns per cycle and that the average downtime per line stop will be 8.0 minutes. The starting blank that is machined on the line costs ₹5.00 per part. The line operates at a cost for 100 parts each and the average cost per tool = ₹.20 per cutting edge. Compute (i) Production rate (ii) Line efficiency (iii) Cost per unit piece produced on the line. (6)

(P.T.O.)

IV. (a) Draw the trajectory of table motion that following NC program seeks to create (4)

N0010G90;
N0011G01X1Y2;
N0012G0IX2Y2;
N0013G91;
N0014G01X1;
N0015G92X2Y2;
N0016G01X1Y1:
N0017G92X0Y0Z0;

(b) Discuss briefly on linear transducer for direct positional measurement. (6)

OR

V. Discuss any ***TWO*** of the following: (2×5=10)

(i) Canned cycles in NC programming with example
(ii) CNC part programming
(iii) Programming in APT

VI. (a) What are the various compensations for machine inaccuracies in CNC? (4)

(b) Discuss briefly different types and its principle of operation on Screw and Nut actuating mechanisms in CNC for converting rotary motion into linear. (6)

OR

VII. Discuss any ***TWO*** of the following: (2 × 5 =10)

(i) Automatic Pallet Changers (APC)
(ii) Automatic Tool changers (ATC)
(iii) Tool Monitoring system (TMS)

VIII. (a) Explain the basic structure of a Robot. (5)

(b) Define flexible manufacturing system, its components, types and applications. (5)

OR

IX. (a) Discuss resolution, accuracy and repeatability in the context of Robot Design. (5)

(b) What is Computer Integrated Process Planning. (5)

***